Adib El Dada

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LinkedIn | GitHub | Portfolio

EDUCATION

McMaster University Sept. 2024 – May 2029
Bachelor of Engineering & Management – Software Engineering & Management *Level III, Co-op*

- Program GPA: **3.7/4.0** (10.2/12), **Dean's Honour List** (2025, 2026), **McMaster Award of Excellence**
- Relevant Coursework: Object-Oriented Programming (**A+**), Computer Architecture (**A+**), Digital Systems & Interfacing (**A+**), Software Eng. Practice & Experience (**A-**), Intro to Software Development (**A-**)

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C, C++, SQL, Bash
Frameworks: Spring Boot, React, Next.js, Node.js, Flask, Tailwind CSS, WebSockets, REST APIs, JWT Authentication
Tools: Git, GitHub Actions, Docker, PostgreSQL, Linux, Maven, JUnit, SonarQube, Vercel

PROJECTS

KeyTrace | *Java, Spring Boot, WebSockets, React, TypeScript* 2026 – Present

- Building a **collaborative coding-interview practice platform**: users share a live **Monaco** editor in rooms joined by link, with every edit relayed over **WebSockets** by a **Spring Boot** backend
- Records **every edit as a timestamped snapshot** and replays the session on a timeline with **play/pause and scrubbing**, so users can see where they got stuck and what they rewrote
- Added **solo, interview and group modes** with per-mode player limits; next: sandboxed code execution in **Docker**, **PostgreSQL** persistence, and **JUnit** tests in CI

RouteScore | *Python, GTFS-Realtime, AWS* 2026 – Present

- Building a tool that measures **how reliable Hamilton (HSR) buses are**, starting with a Python ingester for the city's live **GTFS-Realtime** trip-updates feed
- Designing it as a **serverless AWS pipeline** (Lambda on an EventBridge schedule, S3 snapshots, DynamoDB, CDK) that classifies arrivals with the city's on-time rule and reports **on-time % by route, stop, and hour**

TriageFlow (Mac-A-Thon) | *TypeScript, Next.js, PostgreSQL, Docker, Vercel* Feb. 2026

- Built backend **Next.js API routes** and managed the **PostgreSQL** database (triage sessions, chat transcripts) for an **AI-assisted hospital intake app**, running Postgres in **Docker**
- Deployed to **Vercel** with **dynamically rendered** API routes, so the staff queue (ranked by urgency, then arrival) always shows live data; an **LLM (Llama 3.3 via Groq)** summarizes intake chats for staff to confirm

Catan Game Simulator | *Java, Maven, JUnit, SonarQube, GitHub Actions* Jan. 2026 – Mar. 2026

- **Top contributor (90 of 128 commits)** on a 4-person team building a turn-based Catan engine with **human and rule-based AI players**
- Implemented **undo/redo** with the **Command pattern** (two history stacks), AI turn logic with **Template Method**, and move scoring with the **Visitor pattern**
- Wrote a **regex command parser** and set up the **JUnit** suite (**40 tests**, incl. boundary cases) with **SonarQube** analysis in **GitHub Actions** CI

BridgeAid – 4th Place, McMaster Engineering Competition | *Python, Flask, React, Leaflet.js* Nov. 2025

- Built the **React** frontend and **live incident map** (**Leaflet.js**, OpenStreetMap) for a community disaster-response platform, and worked on the **Python (Flask)** backend secured with **JWT authentication**
- Incidents appear only after **3 unique users** report within **200 m** (haversine distance), filtering out false and spam reports

EXPERIENCE

Tuffee May 2025 – Aug. 2025
Customer Experience Associate

- Handled front-line service at an **industrial supplier of hydraulic and pneumatic parts**, resolving customer inquiries and order issues under time pressure
- Ran **point-of-sale transactions**, **inventory checks**, and **opening/closing procedures** as part of a small team